

Developer Guide

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Adobe

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ISE Boilerplate - Developer Guide

A complete technical guide for developers maintaining and extending this AEM Edge Delivery Services project. The project is built on the AEM Block Collection / aem-boilerplate, authored through Document Authoring (DA) and Universal Editor (UE), with no build step and zero runtime dependencies.

Quick Reference

Resource	URL
Code Repository	github.com/aemdemos/ise-boilerplate
Preview	https://main-ise-boilerplate-aemdemos.aem.page/
Live	https://main-ise-boilerplate-aemdemos.aem.live/
Feature Branch	https://{branch}-ise-boilerplate-aemdemos.aem.page/
Local Dev	http://localhost:3000
Config Service Org	aemdemos

Architecture Overview

Tech Stack

- Vanilla JavaScript (ES6+ native modules) — no transpiling, no bundler
- Native CSS3 with custom properties (design tokens) — no preprocessors or frameworks
- No build step for runtime code — files are served directly to the browser
- Zero production dependencies (devDependencies only, for linting/testing/release)
- Content authored in Document Authoring (da.live) and edited in Universal Editor
- Node.js 24 for local tooling; npm only

Project Structure

```
├── blocks/                # 20 blocks, each {name}/{name}.js + {name}/{name}.css
├── scripts/
│   ├── aem.js             # Core AEM library (DO NOT MODIFY)
│   ├── scripts.js         # Page decoration entry point + E-L-D orchestration
│   ├── delayed.js         # Delayed-phase functionality (analytics/martech)
│   ├── utils.js           # Shared helpers (YouTube/Vimeo embed builders)
│   ├── slider.js          # Shared carousel/slider engine
│   ├── dompurify.min.js   # HTML sanitizer (lazy-loaded on demand)
│   ├── editor-support.js  # Universal Editor live-editing support
│   └── editor-support-rte.js # Rich-text editing support
├── styles/
│   ├── styles.css         # LCP-critical global styles + design tokens
│   ├── lazy-styles.css    # Below-the-fold global styles
│   └── fonts.css          # @font-face declarations (Roboto)
└── ue/
    └── models/            # SOURCE of truth for UE model JSON (see "UE Models")
```

```

├── blocks/      # Per-block component definition/model/filter fragments
├── scripts/     # ue.js + ue-utils.js (UE-only runtime support)
├── icons/       # SVG icons (search.svg)
├── fonts/       # Roboto woff2 web fonts
├── tools/       # Sidekick + quick-edit tooling
├── tests/       # Accessibility tests (Playwright + axe-core)
├── component-definition.json # GENERATED — merged from ue/models/
├── component-models.json    # GENERATED — merged from ue/models/
├── component-filters.json   # GENERATED — merged from ue/models/
├── head.html              # Global <head> content (DO NOT MODIFY)
└── 404.html              # Custom error page

```

Three-Phase Loading (Eager - Lazy - Delayed)

AEM Edge Delivery Services uses a strict Eager-Lazy-Delayed (E-L-D) loading strategy to achieve a Lighthouse score of 100. The orchestration lives in `scripts/scripts.js` via `loadEager()`, `loadLazy()`, and `loadDelayed()`, all driven by `loadPage()`.

Phase	Function	What Loads	Performance Impact
Eager	<code>loadEager(doc)</code>	<code>styles/styles.css</code> , decorates <code><main></code> (<code>decorateMain</code>), loads the first section, waits for the first image (LCP). Fonts loaded eagerly only on desktop (<code>window.innerWidth >= 900</code>) or if already cached in <code>sessionStorage</code> .	Blocks LCP — kept minimal
Lazy	<code>loadLazy(doc)</code>	Remaining sections (<code>loadSections</code>), header, footer, <code>lazy-styles.css</code> , <code>fonts.css</code> , modal autolinking, hash-scroll, Sidekick quick-edit listeners.	Runs after first paint — safe for non-critical UI
Delayed	<code>loadDelayed()</code>	Dynamically imports <code>scripts/delayed.js</code> via <code>requestIdleCallback</code> (3s timeout fallback). Intended for analytics, martech, chat widgets.	Runs ~3s after load — never blocks rendering

Rules for developers:

- Never add third-party scripts to `scripts.js` — they block LCP. Use `delayed.js`.
- `delayed.js` currently contains no integrations (placeholder comment only) — add analytics/martech here.
- Fonts are loaded lazily (or conditionally eager on desktop) to avoid render-blocking; never force eager font loading.
- The first section's blocks load eagerly; all others lazily, automatically based on DOM position.
- Header and footer load in the lazy phase via `loadHeader()` / `loadFooter()` from `aem.js`.
- `loadDelayed()` uses `requestIdleCallback` to avoid INP/TBT regressions.

Key Files and How They Connect

File	Role	Connects To
<code>scripts/aem.js</code>	Core AEM library — <code>loadBlock</code> , <code>loadCSS</code> ,	Imported by <code>scripts.js</code> and most blocks

File	Role	Connects To
	loadScript, decorateBlocks, decorateIcons, loadSection(s), loadHeader/ loadFooter, readBlockConfig, toClassName, getMetadata, createOptimizedPicture, waitForFirstImage, decorateTemplateAndTheme, and the shared DOMPURIFY config. DO NOT MODIFY.	
scripts/scripts.js	Entry point. Orchestrates E-L-D, runs decorateMain (icons, auto-blocks, sections, blocks, buttons, a11y links). Exports project helpers: moveInstrumentation / moveAttributes (UE), getBlockId, decorateButtons, decorateSections, ensureDOMPurify, NX_ORIGIN. Handles section metadata including custom background-color / background-image decorations and style class mapping.	Imports from aem.js; imported by blocks for moveInstrumentation, getBlockId, ensureDOMPurify
scripts/delayed.js	Delayed-phase module (currently empty). Add analytics/marketing tags here.	Dynamically imported by loadDelayed()
scripts/utils.js	Shared YouTube/Vimeo embed-HTML builders (getYoutubeEmbedHtml, getVimeoEmbedHtml) with autoplay/background support.	Imported by video and embed blocks
scripts/slider.js	Shared slider engine — createSliderControls, initSlider, showSlide.	Imported by carousel and card-carousel blocks; showSlide re-exported for UE
scripts/dompurify.min.js	HTML sanitizer used to mitigate DOM XSS. Loaded once on demand via ensureDOMPurify().	Used by quote, table, video, embed, fragment
styles/styles.css	LCP-critical CSS + design tokens (CSS custom properties on :root). Loaded eagerly.	Referenced by all blocks/pages
styles/fonts.css	@font-face declarations for Roboto. Loaded lazily.	Font families referenced in styles.css
styles/lazy-styles.css	Non-critical global styles, loaded lazily.	Supplements styles.css
blocks/{name}/{name}.js	Block logic — exports default function decorate(block). Auto-	May import aem.js, scripts.js helpers, utils.js, slider.js

File	Role	Connects To
	loaded when the block class appears in the DOM.	
blocks/{name}/ {name}.css	Block styles, auto-loaded with the block JS. Some blocks (e.g. hero) split tokens into {name}-tokens.css.	Uses tokens from styles.css
ue/scripts/ue.js	Universal Editor runtime support, dynamically imported in scripts.js only when host is ue.da.live. Handles tab/slide resync and UE instrumentation.	Imports slider.js, ue-utils.js, lazily tabs.js

Execution flow: page load → scripts.js → (if ue.da.live, import ue/scripts/ue.js) → loadPage() → loadEager() (first section + LCP image) → loadLazy() (remaining sections, header, footer, lazy CSS, fonts) → loadDelayed() (imports delayed.js) → loadSidekick().

Universal Editor and Document Authoring Support

This project supports both authoring surfaces:

- **Document Authoring (DA)** — content authored on da.live; scripts.js includes DA-specific handling (e.g. richtext wrapping in decorateSections, the DA Sidekick loaded via loadSidekick, and DA NX preview/experiment hooks reading the nx/dapreview/daexperiment query params).
- **Universal Editor (UE)** — when the page is opened from ue.da.live, scripts.js dynamically imports ue/scripts/ue.js before loadPage(). The project exposes moveInstrumentation(from, to) (and moveAttributes) so blocks preserve data-aue-* / data-richtext-* instrumentation attributes when they rebuild DOM. Most blocks call moveInstrumentation during decoration so authors can edit them in-context.

UE relies on three generated JSON files at the repo root (component-definition.json, component-models.json, component-filters.json). These are NOT edited by hand — see “Universal Editor Models” below.

Local Development Setup

Prerequisites

- Node.js 24
- AEM CLI: `npm install -g @adobe/aem-cli`

Setup Steps

```
git clone https://github.com/aemdemos/ise-boilerplate.git
cd ise-boilerplate
npm install          # also runs `playwright install chromium` via postinstall
aem up              # or: npx -y @adobe/aem-cli up --no-open --forward-browser-logs
```

- Local server runs at `http://localhost:3000` with auto-reload.
- Inspect delivered HTML/DOM with `curl http://localhost:3000/{path}` (or append `.plain.html`).

Test Content (without authored content)

Create static HTML files in a `drafts/` folder and start the dev server with `--html-folder drafts`. The `content/` folder in this repo contains several `*.plain.html` fixtures (e.g. `index.plain.html`, `test.plain.html`, `section-title-test.plain.html`) useful for local rendering.

Linting

```
npm run lint      # runs lint:js then lint:css
npm run lint:js   # eslint . --ext .json,.js,.mjs
npm run lint:css  # stylelint blocks/**/*.css styles/*.css
npm run fix-js    # eslint --fix
npm run fix-css   # stylelint --fix
```

ESLint uses `eslint-config-airbnb-base` plus security/quality plugins (`sonarjs`, `secure-coding`, `browser-security`, `xwalk`). Stylelint uses `stylelint-config-standard`. Linting runs automatically on every PR via GitHub Actions and via a Husky pre-commit hook.

Accessibility Tests

```
npm run test:a11y # node tests/a11y/run.js (Playwright + @axe-core/playwright)
```

Design System

The design system uses a two-tier token model:

- **Tier 1** — global semantic tokens declared in `:root` of `styles/styles.css` (the surface a theme/customer overrides).
- **Tier 2** — component tokens in `blocks/{name}/{name}-tokens.css` that reference Tier 1 tokens (e.g. `hero-tokens.css`).

Migration marker: the boilerplate carries a trailing `/* @boilerplate */` comment on every still-untouched CSS line. As styling is applied for a customer, each edited line's marker is deleted (never flipped). A Claude Code hook (`.claude/hooks/check-boilerplate-markers.mjs`) flags lines edited but left marked — but it stays silent in this source repo. Audit remaining work with:

```
grep -rn '/* @boilerplate' styles/ blocks/ # every untouched line
grep -rc '/* @boilerplate' styles/ blocks/ # remaining count per file
```

CSS Custom Properties (Tokens)

Colors:

```
--background-color: white;
--light-color: #f8f8f8;
--dark-color: #505050;
--text-color: #131313;
--link-color: #3b63fb;
--link-hover-color: #1d3ecf;
--overlay-background-color: lightgrey;
--error-color: firebrick;
/* semantic aliases */
--color-surface: var(--background-color);
```

```
--color-on-surface: var(--text-color);
--color-action: var(--link-color);
--color-action-hover: var(--link-hover-color);
```

Typography:

```
--body-font-family: roboto, roboto-fallback, sans-serif;
--heading-font-family: roboto-condensed, roboto-condensed-fallback, sans-serif;
--fixed-font-family: 'Roboto Mono', menlo, consolas, monospace;
--body-line-height: 1.6;
--heading-line-height: 1.25;
--heading-font-weight: 600;
```

Type scale (mobile base; reduced at $\geq 900\text{px}$):

Token	Mobile	Desktop ($\geq 900\text{px}$)
--body-font-size-m	22px	18px
--body-font-size-s	19px	16px
--body-font-size-xs	17px	14px
--heading-font-size-xxl (h1)	55px	45px
--heading-font-size-xl (h2)	44px	36px
--heading-font-size-l (h3)	34px	28px
--heading-font-size-m (h4)	27px	22px
--heading-font-size-s (h5)	24px	20px
--heading-font-size-xs (h6)	22px	18px

Spacing, layout, borders, motion:

```
--spacing-xs: 8px; --spacing-s: 16px; --spacing-m: 24px;
--spacing-l: 32px; --spacing-xl: 40px;
--max-width-site: 1200px; --icon-size: 24px;
--nav-height: 64px; --breadcrumbs-height: 34px; --header-height: var(--nav-height);
--border-width-s: 1px; --border-color: #dadada;
--border-radius-s: 4px; --border-radius-m: 8px;
--transition-duration: 0.2s;
--block-margin-top: 0.8em; --block-margin-bottom: 0.25em;
--focus-ring-width: 2px; --focus-ring-offset: 2px; --focus-ring-color: var(--link-color);
```

Button tokens (`--button-*`) drive the global `a.button` / button styling. `decorateButtons()` in `scripts.js` maps author formatting to variants: `link` $\rightarrow$ `.button.primary`, `link` $\rightarrow$ `.button.secondary`, `link` $\rightarrow$ `.button.accent`.

Fonts

Roboto is the project font, self-hosted as `woff2` in `fonts/` and declared in `styles/fonts.css`:

Family	Weights	Source file	Usage
roboto	400, 500, 700	roboto-regular.woff2, roboto-medium.woff2, roboto-bold.woff2	Body text
roboto-condensed	700	roboto-condensed-bold.woff2	Headings

Fallback `@font-face` rules in `styles.css` (`roboto-fallback`, `roboto-condensed-fallback`) use `size-adjust` against local Arial to minimize CLS. All faces use `font-display: swap`.

Breakpoints

Mobile-first, `min-width` only. The primary breakpoint in this project is **900px** (desktop).

Name	Min-Width	Usage
Mobile	0 (default)	Base styles
Tablet	600px	<code>@media (width >= 600px)</code>
Desktop	900px	<code>@media (width >= 900px)</code> — type-scale shift, section/nav layout
Wide	1200px	<code>--max-width-site</code> content cap

Section Styles

Section styling is driven by section metadata (`style` field) and decorated in `decorateSections()`:

- **light** and **highlight** — both apply `background-color: var(--light-color)` with `--spacing-xl` vertical padding (see `styles.css`).
- **Background color / image** — sections support `background-color` (validated hex/rgb/name/token) and `background-image` (`http(s)` only) metadata; the image is injected as a decorative `.bg-image` layer behind the content. Values are sanitized in `scripts.js` to prevent CSS/URL injection (CWE-770/915 guards).
- Any other comma-separated `style` values are converted to classes via `toClassName` and added to the section element.

Blocks Reference

Twenty blocks ship with the project. Many call `getBlockId()` to assign unique ids (for ARIA `aria-controls/aria-labelledby` and martech tracking) and `moveInstrumentation()` to preserve UE editing attributes. Blocks needing HTML sanitization call `ensureDOMPurify()`.

Block	Purpose	Variants / Options	Key Details
accordion	Expand/collapse list of label + body rows	—	Rebuilds rows into <code></code> of <code>.accordion-item</code> ; toggles <code>.active</code> on label click; preserves UE instrumentation
card	Shared single-card builder (not authored directly)	—	Exports <code>createCard(row)</code> producing a card <code></code> with <code>.cards-card-image</code> / <code>.cards-card-body</code> ; reused by <code>cards</code> and <code>card-carousel</code>
cards	Grid of cards	—	Builds <code></code> of cards; optimizes images via

Block	Purpose	Variants / Options	Key Details
			<code>createOptimizedPicture</code> (750px); ARIA region
<code>card-carousel</code>	Horizontally sliding cards	single-slide auto-detected	Uses <code>sharedSlider.js</code> (<code>createSliderControls</code> , <code>initSlider</code> , <code>showSlide</code>); reuses <code>createCard</code>
<code>carousel</code>	Full-width slide carousel (image + content)	single-slide auto-detected	Uses <code>slider.js</code> ; each slide split into <code>.carousel-slide-image</code> / <code>.carousel-slide-content</code> ; sets <code>aria-labelledby</code> from heading
<code>columns</code>	Multi-column layout	auto class <code>columns-{N}-cols</code>	Adds <code>.columns-img-col</code> to image-only columns; ARIA region
<code>embed</code>	Embeds videos/social posts	<code>embed-youtube</code> , <code>embed-vimeo</code> (auto from URL), <code>embed-is-loaded</code> , <code>embed-placeholder</code>	Lazy-loads on intersection; uses <code>utils.js</code> embed builders; default iframe wrapper for other hosts
<code>footer</code>	Site footer	—	Loads footer fragment from footer metadata path (default / footer) via <code>loadFragment</code>
<code>form</code>	Renders a form from a JSON definition	—	Fetches form JSON, builds fields via <code>form-fields.js</code> , groups <code>fieldsets</code> , sets <code>data-action</code> submit href
<code>fragment</code>	Inline another document as content	—	<code>loadFragment(path)</code> fetches <code>.plain.html</code> , sanitizes via <code>DOMPurify</code> , copies section classes onto host; also drives auto-blocking of / fragments / links
<code>header</code>	Site navigation / header	<code>nav-drop</code> (dropdown sections), <code>nav-hamburger</code> (mobile)	Loads nav fragment from nav metadata; desktop/mobile behavior via <code>matchMedia</code> (900px); ESC + focus-out close handling
<code>hero</code>	Full-bleed hero with background image	CSS-only block	No JS (<code>hero.js</code> is empty); styled via <code>hero.css</code> + <code>hero-tokens.css</code> ; picture positioned behind heading
<code>modal</code>	Modal dialog	—	No decorate; links to / modals/ auto-open via <code>autolinkModals</code> in <code>scripts.js</code> ; exports <code>createModal/openModal</code> ; uses native <code><dialog></code>
<code>quote</code>	Blockquote with attribution	—	<code>.quote-quotation</code> + <code>.quote-attribution</code> ; wraps <code></code> attribution in <code><cite></code> (<code>DOMPurify</code> -sanitized); ARIA region
<code>search</code>	Client-side site search	<code>no-results</code> state	Reads search index, highlights matching terms, optimizes result images, computes heading level dynamically

Block	Purpose	Variants / Options	Key Details
section-title	Heading + optional subtitle	size size-xs...xxl, alignment left/center/right, color section-title-color-{background, light, dark, text, link, link-hover}, subtitle subtitle-size-*	Supports legacy 4-row table and UE/DA key-value rows; color keys map to :root tokens
table	Renders a table	no-header	Builds <table> with optional <thead>; cells sanitized via DOMPurify; ARIA region
tabs	Tabbed panels	—	Tablist with delegated click handling, aria-controls/aria-selected; UE resync via ue.js; container block (tabs-item children)
video	Inline / background video	autoplay, placeholder (auto), background mode	Supports MP4, YouTube, Vimeo (via utils.js); respects prefers-reduced-motion; click-to-play placeholder

Universal Editor Models

The three root JSON files consumed by Universal Editor are **generated**, not hand-edited:

- component-definition.json
- component-models.json
- component-filters.json

Source of Truth: `ue/models/`

Edit the fragments under `ue/models/` and regenerate. The directory contains:

- Top-level fragments: `component-definition.json`, `component-models.json`, `component-filters.json`, plus base models `page.json`, `section.json`, `text.json`, `image.json`.
- `ue/models/blocks/` — one JSON per editable block (accordion, card, card-carousel, cards, carousel, columns, fragment, hero, quote, search, section-title, table, tabs, video). Each may declare definitions, models, and filters (filters used for container blocks such as tabs).

Build Process

`merge-json-cli` merges the fragments into the three root files:

```
npm run build:json
```

This runs three merges in parallel (`npm-run-all -p`):

```
npm run build:json:models      # ue/models/component-models.json    -> component-models.json
npm run build:json:definitions # ue/models/component-definition.json -> component-definition.json
npm run build:json:filters     # ue/models/component-filters.json  -> component-filters.json
```

Workflow to add/modify a UE-editable block:

1. Add the block to a test page and open it in Universal Editor.

2. Inspect the `/details` network call to understand the structure.
3. Create or edit the block's fragment in `ue/models/blocks/{block}.json` (definition, model, and filter if it is a container).
4. Register the block in `ue/models/section.json`'s filter list.
5. Run `npm run build:json` and commit both the `ue/models/` source and the regenerated root JSON.

For block options, use select components named `classes` or `classes_{suffix}`; they are combined into the block's class list at render time.

Common Development Tasks

Add a New Block

1. Create `blocks/{name}/{name}.js` exporting default function `decorate(block)`.
2. Create `blocks/{name}/{name}.css` with all selectors scoped to `.{name}`.
3. Read the DOM the backend delivers, transform it in place, preserve UE instrumentation with `moveInstrumentation` if you rebuild nodes.
4. If the block needs UE editing, add `ue/models/blocks/{name}.json`, register it in `ue/models/section.json`, and run `npm run build:json`.
5. Test locally at `http://localhost:3000`.

Modify Global Styles

1. Edit `styles/styles.css` (tokens) or `styles/lazy-styles.css` (non-critical).
2. Use `var(--token)` values; remove the `/* @boilerplate */` marker on any line you intentionally change.
3. Test across multiple pages and watch for CLS.

Add Analytics / Marketing Tool

1. Add the integration to `scripts/delayed.js` (currently empty).
2. Never add third-party scripts to `scripts.js` — it blocks LCP.
3. Verify in the Network tab that it loads ~3s after page load.

Debug a Block

1. Check the browser console for errors.
2. Compare the expected vs actual DOM (`console.log(block.innerHTML)`).
3. Confirm variant classes are applied (e.g. `embed-youtube`, `no-header`, `columns-3-cols`).
4. Check CSS specificity and that the block CSS file loaded (Network tab).

Environments

Environment	URL Pattern	Purpose
Local	<code>http://localhost:3000</code>	Development
Feature Branch	<code>https://{branch}-ise-boilerplate-aemdemos.aem.page/</code>	PR testing
Preview	<code>https://main-ise-boilerplate-aemdemos.aem.page/</code>	Staging

Environment	URL Pattern	Purpose
Live	<code>https://main-ise-boilerplate-aemdemos.aem.live/</code>	Production

URL parts: {repo} = ise-boilerplate, {owner} = aemdemos; {branch} from git branch.

Git and Release Workflow

- Branch naming: feature/description, fix/description (keep short for URL limits).
 - Commits follow Conventional Commits — releases are automated with semantic-release (`.releaserc.cjs`); `npm run release` runs in CI.
 - Husky pre-commit hook runs linting; CI runs `npm run lint` on every PR.
 - PR requirements: include before/after preview URLs, pass linting, no console errors, PageSpeed Performance must score 100 on the preview URL. Run `gh pr checks` before requesting review.
-

Troubleshooting

Block not loading

1. Confirm the block folder name matches the authored block class name.
2. Verify the JS exports default function `decorate(block)` (note: `modal` and `card` are intentional exceptions — `modal` auto-links, `card` exports `createCard`).
3. Check the Network tab for 404s on `blocks/{name}/{name}.js / .css`.

Styles not applying

1. Check CSS specificity and that selectors are scoped to the block.
2. Confirm the file loaded (Network tab).
3. Look for stray `/* @boilerplate */`-marked lines you meant to change.

Universal Editor changes not appearing

1. Did you edit `ue/models/` and run `npm run build:json`? The root JSON files are generated.
2. Confirm the block is registered in `ue/models/section.json`'s filter list.
3. Ensure the block preserves instrumentation with `moveInstrumentation` when it rebuilds DOM.

Content not updating

1. Clear browser cache.
2. Confirm preview was triggered in Document Authoring / Sidekick.
3. Allow 1-2 minutes for CDN cache.

Sanitization / embed issues

- `video`, `embed`, `quote`, `table`, and `fragment` depend on DOMPurify via `ensureDOMPurify()`; if content renders empty, check the console for sanitizer-stripped markup.
- YouTube/Vimeo logic lives in `scripts/utls.js` — autoplay and background modes are controlled by the `autoplay` block class and URL parameters.

Resources

Resource	URL
AEM Edge Delivery Services Docs	https://www.aem.live/docs/
Developer / UE Tutorial	https://www.aem.live/developer/ue-tutorial
Universal Editor Blocks	https://www.aem.live/developer/universal-editor-blocks
Component Model Definitions	https://www.aem.live/developer/component-model-definitions
Block Collection	https://www.aem.live/developer/block-collection
Keeping It 100 (E-L-D / performance)	https://www.aem.live/developer/keeping-it-100
Markup, Sections, Blocks, Auto Blocking	https://www.aem.live/developer/markup-sections-blocks
Universal Editor (da-live wiki)	https://github.com/adobe/da-live/wiki/Universal-Editor